

Busy Snail Filament Changer (BSFC) BOM V.2026-02-03 for 4x1kg spool units changer, totals

	Item	Qty	Info	2026-02 sample link
	Heat set insert M3xD5xL4	68	Same as Voron BOM size	
	Flat Head Cap Screw M3x6	4		
	Flat Head Cap Screw M3x8	24		
	Socket Head Cap Screw M3x6	8		
	Socket Head Cap Screw M3x10	8		
	Socket Head Cap Screw M3x14	4		
	Socket Head Cap Screw M3x20	16		
	Socket Head Cap Screw M3x25	4		
	Socket Head Cap Screw M3x30	12		
	Self tapping screw M2x5	6		
	Self tapping screw M2x8 flat tip	36		<u>SELF TAPPING FLAT TIP</u>
	Nut M12	4	M12x1.75 DIN934	<u>NUT</u>
	Washer M12	4	M12x24x2.5 DIN125	<u>WASHER</u>
	Threaded rod M12	2	M12x1.75 DIN975, length depends on the changer and printer setups	<u>THREADED ROD</u>
	BMG 17 teeth extruder gear	4	Extruder drive gear with grub screw for 5mm stepper shaft	<u>BMG-17 GEAR</u>
	Ball bearing MR63zz	8+1	Included 8 in 4x HGX-lite extruder part kits, 1 extra needed	<u>HGX-LITE KIT / BEARING MR63zz</u>
	Thumb screw assy with spring	4+4	Included 4 in 4x HGX-lite extruder part kits, 4 extras needed	<u>HGX-LITE KIT / THUMB SCREW ASSY</u>
	Large extruder gear	8	Included in 4x HGX-lite extruder part kits	<u>HGX-LITE KIT</u>
	Needle bearing 3x5x6.5mm	8	Included in 4x HGX-lite extruder part kits	<u>HGX-LITE KIT</u>
	Bearing pin 3x16mm	8	Included in 4x HGX-lite extruder part kits	<u>HGX-LITE KIT</u>
	Bearing pin 3x12mm	12		<u>BEARING PIN</u>
	Tension spring D6xL20xWD0.4	1	Tension spring for buffer slide (Total length with eyelets 6+8+6=20mm)	<u>TENSION SPRING</u>
	PTFE tube OD4mm ID2mm	1m		<u>PTFE TUBE</u>
	PTFE tube OD6mm ID4mm	3m	3 meter continuous lot	<u>PTFE TUBE</u>
	Tube collet 4mm	5		<u>TUBE COLLET 4mm</u>
	Tube collet 6mm	9		<u>TUBE COLLET 6mm</u>
	Cable ties, small	20	1.8mm tie width, 3.5mm eye width	<u>SMALL CABLE TIE</u>
	Bicycle inner tube	≈0.3m	Optimal flat width ≈40-45mm to be stretched on a 32mm diameter roller, 8x35mm pieces	

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	Stepper motor NEMA17 L=38mm	4	Optimal L=38(40)mm can be longer but affects wiring length and units sideways stacking fitment	NEMA17
	Led SK6182 RGBW	4		LED
	Micro switch D2F	9		MICRO SWITCH
	Micro switch D2F-L	2		MICRO SWITCH
	JST PH2.0 3-m-pin housing	12	Switch and led inlet drive unit	JST PH2.0 HOUSING
	JST PH2.0 4-m-pin housing	5	Motor inlet drive unit (4) and hub/buffer (1)	JST PH2.0 HOUSING
	JST PH2.0 6-fm-pin housing	4	Stepper motor connector	JST PH2.0 HOUSING
	JST PH2.0 3-fm-pin housing	12	Wirings on drive unit ends	JST PH2.0 HOUSING
	JST PH2.0 4-fm-pin housing	5	Wirings on drive unit and hub/buffer ends	JST PH2.0 HOUSING
	JST PH2.0 male pin	56		JST PH2.0 MALE PIN
	JST PH2.0 female pin	72		JST PH2.0 FEMALE PIN
	FEP wire UL1332 26AWG	?	OD max 1.1mm to fit wire bundles to channels in 3d printed parts at some tight places	UL1332 FEP WIRE
	Controller board*	1	For 4 spool units changer: a board with 4 stepper drivers, minimum 11 switch inputs and rgb	BTT MMB CAN V2

Controller board box supports Btt Mmb Can V1 and v2 boards. Wiring connectors on board end are not included in this BOM.